```
    project_src: "{{ playbook_dir }}/."
    state: absent
```

The above playbook can be invoked using the following command:

```
$ sudo ansible-playbook provision.yml --tags stop
```

You can also verify that there are no containers running in the system, as follows:

```
$ docker ps
CONTAINER ID    IMAGE         COMMAND             CREATED
STATUS          PORTS         NAMES
```

Refer to the Ansible *docker_service* module's documentation at *http://docs.ansible.com/ansible/ latest/ docker_service_module.html* for more examples and options.

Vagrant and Ansible

Vagrant is free and open source software (FOSS) that helps to build and manage virtual machines. It allows you to create machines using different backend providers such as VirtualBox, Docker, libvirt, etc. It is developed by HashiCorp and is written in the Ruby programming language. It was first released in 2010 under an MIT licence. The Vagrantfile describes the virtual machine using a Ruby DSL, and an Ansible playbook can be executed as part of the provisioning process.

The following dependencies need to be installed on the host Ubuntu system:

```
$ sudo apt-get build-dep vagrant ruby-libvirt
$ sudo apt-get install qemu libvirt-bin ebtables dnsmasq
virt-manager
$ sudo apt-get libxslt-dev libxml2-dev libvirt-dev zlib1g-dev
ruby-dev
```

Vagrant 1.8.7 is then installed on Ubuntu using a *.deb* package obtained from the *https://www.vagrantup.com/* website. Issue the following command to install the *vagrant-libvirt* provider:

```
$ vagrant plugin install vagrant-libvirt
```

The *firewalld* daemon is then started on the host system, as follows:

```
$ sudo systemctl start firewalld
```

A simple Vagrantfile is created inside a test directory to launch an Ubuntu guest system. Its file contents are given below for reference:

```
# -*- mode: ruby -*-
# vi: set ft=ruby :

Vagrant.configure("2") do |config|
  config.vm.define :test_vm do |test_vm|
    test_vm.vm.box = "sergk/xenial64-minimal-libvirt"
  end

  config.vm.provision "ansible" do |ansible|
    ansible.playbook = "playbook.yml"
  end
end
```

When provisioning the above Vagrantfile, a minimalistic Xenial 64-bit Ubuntu image is downloaded, started and the Ansible playbook is executed after the instance is launched. The contents of the *playbook.yml* file are as follows:

```
---
- hosts: all
  become: true
  gather_facts: no

  pre_tasks:
    - name: Install python2
      raw: sudo apt-get -y install python-simplejson

  tasks:
    - name: Update apt cache
      apt: update_cache=yes

    - name: Install Apache
      apt: name=apache2 state=present
```

The minimal Ubuntu machine has Python 3 by default, and the Ansible that we use requires Python 2. Hence, we install Python 2, update the APT repository and install the Apache Web server. A sample debug execution of the above playbook from the test directory is given below:

```
$ VAGRANT_LOG=debug sudo vagrant up --provider=libvirt

Bringing machine 'test_vm' up with 'libvirt' provider...
==> test_vm: Creating image (snapshot of base box volume).
==> test_vm: Creating domain with the following settings...
==> test_vm:  -- Name:              vagrant-libvirt-test_
test_vm
==> test_vm:  -- Domain type:       kvm
==> test_vm:  -- Cpus:              1
==> test_vm:  -- Feature:           acpi
==> test_vm:  -- Feature:           apic
==> test_vm:  -- Feature:           pae
==> test_vm:  -- Memory:            512M
==> test_vm:  -- Management MAC:
==> test_vm:  -- Loader:
```